

How to Read Study Results | ClinicalTrials.gov

Learn how to read study results in records posted on ClinicalTrials.gov. Results include information about the study's participant flow, baseline characteristics, outcome measures, and adverse events (including serious adverse events).

The study results provided in a ClinicalTrials.gov record are not meant to be interpreted without help from an expert. Interpreting study results is complex because many factors can influence their meaning. For more information about the results of a specific study, contact the Results Point of Contact found in the Results Overview section.

You can search for clinical studies with results by using filters on the ClinicalTrials.gov home page or the search results page. See [How to search for studies with results](https://clinicaltrials.gov/find-studies/how-to-search-for-studies-with-results)(https://clinicaltrials.gov/find-studies/how-to-search-for-studies-with-results).

Overview

The Results Posted tab of a study record includes these 10 main sections:

Results Overview - general information about the study

Study Record Dates - dates for study registration and results reporting

Participant Flow - the progress of participants through the overall study

Baseline Characteristics - the data that researchers collected at the start of the study about all the participants

Outcome Measures - the data collected about participants during the study

Adverse Events - any adverse events that happened to participants during the study

Limitations and Caveats - explanations of anything that might have impacted the quality or reliability of the study results, if provided

Collaborators and Investigators - the people and organizations involved in the study

Publications - links to any published scientific papers related to the study (this section will not be included if there are no publications)

More Information - a link to the study's record history and details about any agreements the study investigators have with the sponsor

Helpful features

Feature	Function
	Click this to see the glossary definition for the term.
+ + Show more	Click these to see more information in a table or a section.
– – Show less	Click these to see less information in a table or a section.
+ Expand all content	Click this to view all information on the page.
– Collapse all content	Click this to view less information on the page.
Expand all	Click this to view all information in a section.
Collapse all	Click this to view less information in a section.

Where to find results in a study record

When you open a study record, you will see a tab labeled “Results Posted,” “Results Submitted,” or “No Results Posted” located between the Researcher View and Record History tabs in the dark blue bar below the study title.

If the study has a Results Posted tab, the study will have results. You can view the results by clicking on that tab.

Download

Save

+ Expand all content

– Collapse all content

Study Details

Researcher View

Results Posted

Record History

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Study Record Dates

Participant Flow

Baseline Characteristics

Outcome Measures

Adverse Events

Limitations and Caveats

Collaborators and Investigators

Results Overview

Conditions

Chemotherapy-Induced Peripheral Neuropathy

Intervention/Treatment

Device: Placebo Administration

Other: Questionnaire Administration

Device: Transcutaneous Electrical Nerve Stimulation

Other Study ID Numbers

Enrollment (Actual)

151

Study Type

Interventional

A study record with the 'Results Posted' tab selected.

If the study has a Results Submitted tab, it means that results have been submitted to ClinicalTrials.gov but have not yet been posted.

If the study has a No Results Posted tab, it means that no results have been submitted to ClinicalTrials.gov. A study may not have results posted because the study isn't done, the deadline for submitting results has not passed, or because the study is not required to submit results.

Sections of the study results

If results have been posted, you will see these 10 possible sections on the Results Posted tab:

[Results Overview](https://clinicaltrials.gov/study-basics/how-to-read-study-results#results-overview-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#results-overview-section)

[Study Record Dates](https://clinicaltrials.gov/study-basics/how-to-read-study-results#study-record-dates-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#study-record-dates-section)

[Participant Flow](https://clinicaltrials.gov/study-basics/how-to-read-study-results#participant-flow-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#participant-flow-section)

[Baseline Characteristics](https://clinicaltrials.gov/study-basics/how-to-read-study-results#baseline-characteristics-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#baseline-characteristics-section)

[Outcome Measures](https://clinicaltrials.gov/study-basics/how-to-read-study-results#outcome-measures-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#outcome-measures-section)

[Adverse Events](https://clinicaltrials.gov/study-basics/how-to-read-study-results#adverse-events-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#adverse-events-section)

[Limitations and Caveats](https://clinicaltrials.gov/study-basics/how-to-read-study-results#limitations-caveats-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#limitations-caveats-section)

[Collaborators and Investigators](https://clinicaltrials.gov/study-basics/how-to-read-study-results#collaborators-investigators-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#collaborators-investigators-section)

[Publications](https://clinicaltrials.gov/study-basics/how-to-read-study-results#publications-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#publications-section)
(only visible if there are publications)

[More Information](https://clinicaltrials.gov/study-basics/how-to-read-study-results#more-information-section)(https://clinicaltrials.gov/study-basics/how-to-read-study-results#more-information-section)

If you are using a computer or tablet and want to jump to a certain section, click on the section name in the left-sided menu of the results page.

Study Details	Researcher View	Results Posted	Record History
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On this page

Results Overview
Study Record Dates
Participant Flow
Baseline Characteristics
Outcome Measures
Adverse Events
Limitations and Caveats
Collaborators and Investigators
More Information

Results Overview

Conditions ⓘ

Chemotherapy-Induced Peripheral Neuropathy

Intervention/Treatment ⓘ

- Device: Placebo Administration
- Other: Questionnaire Administration
- Device: Transcutaneous Electrical Nerve Stimulation

Other Study ID Numbers ⓘ

- URCC19085

The computer/tablet view of the navigation menu within the 'Results Posted' section of a study record.

If you are using a mobile device such as a smartphone, and want to jump to a certain section, start scrolling down the page. “On this page” will appear in the blue bar at the top. Click on that bar, and a menu showing the sections of the Results Posted tab will appear. Click on a section to go to it or click on the “X” to close the menu.

≡ On this page
X

Results Posted

Results Overview
Study Record Dates
Participant Flow
Baseline Characteristics

Outcome Measures	
Adverse Events	
Limitations and Caveats	
Collaborators and Investigators	 Feedback
More Information	

The mobile device view of the navigation menu within the 'Results Posted' section of a study record.

Results Overview section

This section includes general information about the study:

On the left side of this section you will find:

Conditions - the disease, disorder, syndrome, illness, or injury that was studied

Intervention/Treatment - the drug, device or action that was studied

Other Study ID Numbers - any identification numbers assigned to a study record other than the NCT Number

Study Design - the details about how the study was set up to answer the study questions

Results Point of Contact - the name and contact information for the person who can answer questions about the study results

On the right side of this section, you will find:

Enrollment - the number of participants who were in the study

Study Type - whether the study was interventional (participants were assigned to treatments) or observational (participants were not assigned to treatments)

Study Record Dates section

This section shows important dates about the study record.

Study Registration Dates - when the record was first submitted and first posted on ClinicalTrials.gov

Results Reporting Dates - when the results were first submitted and first posted on ClinicalTrials.gov

Study Record Updates - when the last update was posted on ClinicalTrials.gov and when the researchers last verified that the information is accurate and current

Additional dates may be included in this section.

Participant Flow section

This section shows the progress of participants through the overall study.

At the top are fields for Recruitment Details and Pre-assignment Details. These fields can explain where participants were from and what happened to participants who were enrolled in but did not start the study.

Recruitment Details	[Not Specified]
Pre-assignment Details	7 participants withdrew from the study before randomization to an arm for the following reasons: Became ineligible after registration, Lost to follow-up, Changed their mind.

An example of the 'Recruitment Details' and 'Pre-assignment Details' fields.

If there was more than one arm or group, each one is shown in a different column at the top of the Participant Flow table, like this:

Arm/Group Title	Group I (Active TENS)	Group II (Placebo TENS)
Arm/Group Description	Patients wear an active wireless TENS device 5 hours daily for up to 6 weeks in the absence of unacceptable toxicity.... + Show more	Patients wear a placebo wireless TENS device 5 hours daily for up to 6 weeks in the absence of unacceptable toxicity.... + Show more
Period Title: Overall Study		
Started	72	72
Received Intervention and Completed Baseline Assessments	72	70
Completed	68	62
Not Completed	4	10
Reason Not Completed		

Adverse Event	1	1
Death	1	0
Lost to Follow-up	1	1
Withdrawal by Subject	0	3
App glitch during the study	1	0
Device Distribution Error	0	2
Ineligible after Randomization	0	1
Other Medical Reasons	0	2

Example of a study that had two groups. The groups are represented by separate columns within a table.

Each period of the study has its own table, which shows the progress of participants through certain stages of the study. The dark blue header shows the title for the period.

In the period table, you'll find the number of people in each arm or group who started, completed, and did not complete each period. There may also be rows that provide more information about the participants and that list the reasons participants did not complete the study. For example, the table above shows the participants who "Received Intervention and Completed Baseline Assessments" as well as the reasons that some participants did not complete the study (under the dark gray Reason Not Completed heading row).

To adjust the view of the table:

Click on the minus sign (-) to the right of the Period Title heading to collapse a period table or click on the plus sign (+) to expand the table.

If there was more than one period in the study, you can expand or collapse all the period tables by clicking on **Expand all** or **Collapse all**, found above the blue heading row of the first period table.

Baseline Characteristics section

This section shows the data that researchers collected about all participants at the start of the study. Baseline Characteristics may include:

Age

Sex or Gender

Race

Ethnicity

Study-Specific Measures - the baseline data for the study’s outcome primary measures or baseline data for other measures unique to the study, for example, blood pressure, height, or weight

Region of Enrollment - including the number of participants in different countries who took part in the study

The first table shows the arm or group names, descriptions, and the number of people whose data were collected at the start of the study.

If there was more than one arm or group, each one is shown in a different column in the table. There is also a Total column for participants in all groups, like this:

Arm/Group Title	Group I (Active TENS)	Group II (Placebo TENS)	Total
Arm/Group Description	Patients wear an active wireless TENS device 5 hours daily for up to 6 weeks in the absence of... + Show more	Patients wear a placebo wireless TENS device 5 hours daily for up to 6 weeks in the absence of... + Show more	Total of all reporting groups
Overall Number of Baseline Participants	72	70	142
Baseline Analysis Population Description	[Not Specified]		

Example of the overall baseline participants table in a study that had two groups. Each group is represented by a column. A third column is the total of both groups. There are rows for Group Description, Overall Number of Baseline Participants, and Baseline Analysis Population Description.

Each baseline characteristic has a table that includes the baseline measurement data. Each table has a blue header with the name of the baseline characteristic, the type of data collected, and the unit of measure. For example, in the table below the name of the baseline characteristic is “Age, Continuous,” the type of data is “Mean (Standard Deviation),” and the unit of measure is “years.” This table shows that there were 72 participants in the Group 1 (Active TENS) arm (the same column in the table above), and their average age was 62.6 years, plus or minus 8.6 years.

Expand all / Collapse all

Age, Continuous Mean (Standard Deviation) Unit of measure: years			
Number Analyzed	72 participants	70 participants	142 participants
	62.6 (8.6)	63.3 (10.7)	62.9 (9.7)

Image alt: Example of a specific baseline characteristic table (in this case, 'Age, Continuous') for a study with two groups. Each group is represented by a column. A third column is the total of both groups.

To adjust the view of the table:

Click on the minus sign (-) to the right of the baseline characteristic name (blue row) to collapse the baseline characteristics table or click on the plus sign (+) to expand the table.

You can expand or collapse all baseline characteristics tables by clicking on **Expand all** or **Collapse all**, found above the blue heading row of the first baseline characteristics table.

Outcome Measures section

This section shows the data that researchers collected about participants during the study.

Each Outcome Measure table has a blue heading row with the name of the outcome measure; whether it was a primary, secondary, or other measure; and the timeframe when the measure was taken. Each outcome measure is numbered. The rows below the blue heading row may include descriptions of the outcome measure and the participants who were studied.

1. Chemotherapy-induced Peripheral Neuropathy (CIPN) Symptoms	
Type: Primary Time Frame: 6 weeks after the start of intervention	
Description	Measured by the mean European Organization for Research and Treatment of Cancer-CIPN20 (EORTC-CIPN20). The effects of transcutaneous electrical nerve stimulation (TENS) on CIPN will be estimated using analysis of covariance (ANCOVA). A 20 -item patient self -report tool to assess symptoms and function in the sensory, motor and autonomic domains. Two items, Q49 and Q50, were excluded from the total score calculation. Q49 was relevant only for individuals who could drive, and Q50 was relevant only for men. 0 - 72, a higher score indicates worse neuropathy
Time Frame	6 weeks after the start of intervention
Analysis Population Description	Participants who completed the baseline assessment were included in the analysis (N=141) except one extreme outlier that was removed from the Active TENS group due to data discordance among similar measures at baseline.

Example of an Outcome Measure table showing the outcome measure name across the top, and rows for description, time frame, and analysis population description.

The rest of the table shows the number of participants included in the measure and the data that were collected. If there was more than one arm or group, each one is shown in a different column in the table, like this:

Arm/Group Title	Group I (Active TENS)	Group II (Placebo TENS)
Arm/Group Description	Patients wear an active wireless TENS device 5 hours daily for up to 6 weeks in the absence of unacceptable toxicity.... + Show more	Patients wear a placebo wireless TENS device 5 hours daily for up to 6 weeks in the absence of unacceptable toxicity.... + Show more
Overall Number of Participants Analyzed	71	70

Least Squares Mean (Standard Error) Unit of Measure: score on a scale	32.32 (0.57)	33.37 (0.59)
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Example of an Outcome Measure table from a study with two groups. Each group is represented by a column.

To adjust the view of the table:

Click on the minus sign (-) to the right of the outcome measure name (blue row) to collapse an outcome measure table or click on the plus sign (+) to expand the table.

You can expand or collapse all outcome measure tables by clicking on **Expand all** or **Collapse all**, found above the blue heading row of the first outcome measure table.

Statistical Analysis tables

Some outcome measures include statistical analysis tables, like the one shown below. Statistical analysis involves mathematical formulas and calculations that help researchers make decisions and draw conclusions about the data from a clinical study. These tables will be collapsed by default. Click on the plus sign (+) to the right of the name of the statistical analysis to expand the table.

Statistical Analysis 1: Group I (Active TENS), Group II (Placebo TENS) Superiority ANCOVA	+
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Example of a Statistical Analysis table header with a plus sign icon indicating the table can be expanded.

Adverse Events section

This section summarizes the number and type of Adverse Events that happened to participants in the study, including:

All-Cause Mortality - how many people in each arm died for any reason during the study

Serious Adverse Events - events that caused death or were life-threatening

Other (Not Including Serious) Adverse Events - adverse events that are not considered serious

Adverse events are shown in tables organized by type (named in the blue heading rows). Tables showing general information appear first.

The first table shows the time frame during which the researchers tracked adverse events. There may also be a description of how the information was collected. The next table shows the title of each arm or group in the study and a description of that arm or group. If there was more than one arm or group, each one is shown in a different column in the tables that follow, like this:

Time Frame	The adverse event data were collected over the intervention period. i.e. 6 weeks.
Adverse Event Reporting Description	Adverse events were assessed by phone in weeks 1, 3, and 5 and at the Week 6 visit. At each participant visit and phone call, the site study staff assessed adverse events by asking participants if they have experienced any new symptoms since starting the experimental treatment. They asked specifically if any skin reactions have occurred.

Arm/Group Title	Group I (Active TENS)	Group II (Placebo TENS)
Arm/Group Description	Patients wear an active wireless TENS device 5 hours daily for up to 6 weeks in the absence of unacceptable toxicity.... + Show more	Patients wear a placebo wireless TENS device 5 hours daily for up to 6 weeks in the absence of unacceptable toxicity.... + Show more

Tables showing information about Adverse Events. One shows the time frame and description of how information was collected. The other lists two participant groups in separate columns.

The next table, All-Cause Mortality, shows how many participants died during the study. In this table, and in the ones that follow, the total number of participants affected (who had the adverse event happen to them) and the number at risk (who could have had the adverse event happen) is shown for each arm or group. Here is an example:

[Expand all](#) / [Collapse all](#)

All-Cause Mortality		
Arm/Group Title	Group I (Active TENS)	Group II (Placebo TENS)
	Affected / at Risk (%)	Affected / at Risk (%)
Total	1/72 (1.39%)	0/70 (0.00%)

Example of the expanded All-Cause Mortality table from a study with two groups.

The final two tables show serious and other adverse events. Events in these tables are organized by the organ systems that were affected, shown in dark gray heading rows. The number of participants affected and the number at risk in each arm or group is shown for each event.

Here is an example of a Serious Adverse Events table, where the adverse event Disease Progression is listed under the General disorders heading (gray row):

Serious Adverse Events				
Arm/Group Title	Group I (Active TENS)		Group II (Placebo TENS)	
	Affected / at Risk (%)	# Events	Affected / at Risk (%)	# Events
Total	1/72 (1.39%)		1/70 (1.43%)	
General disorders				
Disease Progression [†]	1/72 (1.39%)	1	0/70 (0.00%)	0
Infections and infestations				

Infections and infestations				
Urinary tract infection [†]	0/72 (0.00%)	0	1/70 (1.43%)	1
Metabolism and nutrition disorders				
Dehydration [†]	0/72 (0.00%)	0	1/70 (1.43%)	1
† Indicates events were collected by systematic assessment				

The Serious Adverse Events table from a study with two groups. Each group is represented by a column. Each group column is divided into two more columns that show the percentage of participants affected and at-risk for the adverse events. Adverse events are listed in the table rows.

In the Other (Not Including Serious) Adverse Events table, the first row shows the frequency threshold for reporting other adverse events, a number that ranges from 0% to 5%. If the threshold is 0%, it means that the table includes *all* other adverse events.

Here is an example:

Other (Not Including Serious) Adverse Events				
Frequency Threshold for Reporting Other Adverse Events	0%			
Arm/Group Title	Group I (Active TENS)		Group II (Placebo TENS)	
	Affected / at Risk (%)	# Events	Affected / at Risk (%)	# Events
Total	17/72 (23.61%)		12/70 (17.14%)	
Injury, poisoning and procedural complications				
Bruising [†]	1/72 (1.39%)	1	0/70 (0.00%)	0
Fracture [†]	0/72 (0.00%)	0	1/70 (1.43%)	1
Metabolism and nutrition disorders				
Hyperglycemia [†]	0/72 (0.00%)	0	1/70 (1.43%)	1
Musculoskeletal and connective tissue disorders				
Muscle cramp [†]	5/72 (6.94%)	5	2/70 (2.86%)	2
Arthritis [†]	0/72 (0.00%)	0	1/70 (1.43%)	1
Myalgia [†]	0/72 (0.00%)	0	1/70 (1.43%)	1
Pain in extremity [†]	1/72 (1.39%)	1	1/70 (1.43%)	1

Other Adverse Events table with a zero percent threshold from a study with two groups. Each group column is further divided into two more columns that show the percentage of participants affected and at-risk for the adverse events. Other adverse events are listed in the table rows.

If the threshold is greater than 0%, it means that the table includes only adverse events that happened

at or above that percentage of participants in any arm or group. Non-serious adverse events that occur at a frequency of 5% or greater in any single arm or group must be included in the Other (Not Including Serious) Adverse Events table.

To adjust the view of the table:

Click on the minus sign (-) to the right of the type of adverse event (blue row) to collapse an adverse event table or click on the plus sign (+) to expand the table.

You can expand or collapse all adverse event tables by clicking on **Expand all** or **Collapse all**, found above the blue heading row of the All-Cause Mortality table.

Limitations and Caveats section

This section may show any limitations or caveats the study had. Limitations and Caveats may explain:

Any events that could have affected the quality of the results

Why a study was not completed as planned

Why recruitment or enrollment ended early

Collaborators and Investigators section

This section shows the people and organizations involved in the study. There are three parts:

Sponsor - the organization or person who controls the study

Collaborators - other organizations that helped carry out the study

Investigators - the researchers who conducted the study

Publications section

This section shows publications (papers in scientific journals) related to the study, if any are available.

This section has three parts:

General - publications listed in the study record by the person who entered information about the study. These publications can be about anything related to the study.

Study Results - publications about the results of the study listed in the study record by the person who entered information about the study.

From PubMed - publications automatically gathered from PubMed, a public database of scientific and medical articles, through a search of PubMed for the ClinicalTrials.gov Identifier (NCT Number). These articles may or may not be about the study.

If there are no publications, this section won't be shown.

More Information section

This section has two parts:

A **Record History** button - goes to the study's Record History tab. This tab shows the dates that information about the study was updated or changed, and it lists the changes that were made. Each version of the record can be viewed, and two versions can be compared.

Certain Agreements - lists agreements between the study sponsor and main researcher (principal investigator) about publishing results after the study ends